Block Blast Pro — Developer Setup Guide

A React Native (Expo SDK 52) block-puzzle game for Android.

This guide takes a new machine from nothing to the app running on a phone. **Follow the steps in order.** Section 7 lists the traps that have already cost this project days of debugging — read it before you file a bug.

---

1. Tech stack

Component	Version	Notes
Expo SDK	52	`expo@~52.0.0`
React Native	0.76.9	Pinned exactly — do not bump casually
React	18.3.1	
Node.js	20 LTS or 22 LTS	24 works but is newer than Expo 52 targets
JDK	**17**	Mandatory. 21/24/25 will NOT work
Android compileSdk	35	
Android targetSdk	34	
Android minSdk	24	RN 0.76 native libs are built for API 24
Build Tools	35.0.0	
NDK	26.1.10909125	Exact version required
Gradle	8.10.2	Provided by the wrapper — don't install manually

---

2. Prerequisites

2.1 Node.js

Install Node **20 LTS** or **22 LTS** from https://nodejs.org.

```
node -v    # v20.x or v22.x
npm -v
```

2.2 JDK 17 — the most common source of failure

Gradle and the Android Gradle Plugin require **JDK 17**. A newer system JDK is the single most frequent cause of a broken build here.

Install Eclipse Temurin JDK 17: https://adoptium.net/temurin/releases/?version=17

Then set `JAVA_HOME` **permanently** (Windows):

```
# PowerShell, run once. Adjust the path to your install.
[Environment]::SetEnvironmentVariable(
  "JAVA_HOME",
  "C:\Program Files\Eclipse Adoptium\jdk-17.0.13.11-hotspot",
  "User")
```

macOS / Linux — add to `~/.zshrc` or `~/.bashrc`:

```
export JAVA_HOME=/Library/Java/JavaVirtualMachines/temurin-17.jdk/Contents/Home
export PATH="$JAVA_HOME/bin:$PATH"
```

Open a **new** terminal and verify:

```
java -version    # must print 17.x
echo $JAVA_HOME  # must not be empty
```

> Do **not** put `org.gradle.java.home` in `android/gradle.properties`. It hardcodes a path containing your username and breaks the build for everyone else.

2.3 Android Studio + SDK

Install Android Studio: https://developer.android.com/studio

In **Settings** → **Languages & Frameworks** → **Android SDK**:

SDK Platforms tab — check *Show Package Details*:

- Android 15 (API 35) — `compileSdk`
- Android 14 (API 34) — `targetSdk`

SDK Tools tab — check *Show Package Details*:

- Android SDK Build-Tools **35.0.0**
- **NDK (Side by side)** → **26.1.10909125** ← exact version, required for native code
- CMake
- Android SDK Platform-Tools
- Android Emulator (skip if you only use a physical device)

2.4 ANDROID_HOME

Windows (PowerShell, once):

```
[Environment]::SetEnvironmentVariable(
  "ANDROID_HOME", "$env:LOCALAPPDATA\Android\Sdk", "User")
```

macOS / Linux:

```
export ANDROID_HOME=$HOME/Library/Android/sdk    # macOS
export ANDROID_HOME=$HOME/Android/Sdk            # Linux
export PATH="$ANDROID_HOME/platform-tools:$ANDROID_HOME/emulator:$PATH"
```

New terminal, then verify:

```
adb --version
```

2.5 Disk space

Budget **25 GB free** before your first build:

Item	Size
Android SDK + NDK	~8 GB
Gradle caches (`~/.gradle`)	~5 GB
One emulator image	~2.5 GB
`node_modules`	~500 MB
Debug APK + build output	~1 GB

A single build extracts a 206 MB React Native archive into a ~2.5 GB Gradle transform cache. Building on a nearly-full disk fails in confusing ways.

---

3. Clone and install

```
git clone <REPO_URL>
cd block-blast-game
npm install
```

> **Never copy `node_modules` from another machine** (no zips, no USB, no network share). It caches absolute paths from the original machine; the native build then fails with `[CXX1210] No compatible library found` . Always `npm install` .

`android/local.properties` is gitignored and generated automatically. If the SDK isn't found, create it manually:

```
# Windows (note the doubled backslashes)
sdk.dir=C:\\Users\\YOUR_NAME\\AppData\\Local\\Android\\Sdk
# macOS
sdk.dir=/Users/YOUR_NAME/Library/Android/sdk
```

---

4. Pick a device

Option A — physical phone (recommended: faster, real performance)

1. **Settings** → **About phone** → tap **Build number** 7 times to enable Developer options.
2. **Settings** → **System** → **Developer options** → enable **USB debugging**.
3. Connect by USB and accept the *Allow USB debugging* prompt on the phone.

```
adb devices     # your device should be listed as "device", not "unauthorized"
```

Option B — emulator

Android Studio → **Device Manager** → **Create Virtual Device** → Pixel 7, system image API 36. Start it, then confirm with `adb devices`.

---

5. Run the app

```
npx expo run:android
```

First run takes **5–15 minutes** (it compiles native code). Later runs are far faster. This command builds the APK, installs it, starts Metro, and launches the app.

Once it's running, **JS and component edits hot-reload instantly** — just save.

When you must rebuild natively

A JS reload can't pick up native code. Re-run `npx expo run:android` after:

- adding/removing any package with a native module (`expo-av`, `expo-haptics`, `@react-native-async-storage/async-storage`, `netinfo`, ...)
- changing anything in `android/`
- editing `app.json` permissions or config plugins

Rule of thumb: **new package → rebuild. JS-only change → hot reload.**

Useful commands

```
npx expo start              # Metro only (JS changes, app already installed)
npx expo start --clear      # Metro with a cleared cache
adb devices                 # list connected devices
adb logcat | grep -i reactnativejs        # JS logs / errors
npx expo install <package>  # add a package at the SDK-52-correct version
```

Use `npx expo install`, **not** `npm install`, for Expo packages — it picks the version matching SDK 52. Plain `npm install` can pull an SDK 53 build that fails to compile.

---

6. Project layout

```
App.js                      Root component, state, persistence hydration
index.js                    Entry point (registerRootComponent)
app.json                    Expo config: name, icons, permissions, AdMob ID
src/
  screens/
    MainMenuScreen.js       Home: play, stats, coins, themes
    GameScreen.js           Board, scoring, combos, game-over
  components/
    GridBoard.js            8x8 grid
    PieceTray.js            Draggable piece tray
    Header.js               Score / coins / controls
    ComboOverlay.js         Combo animation
    BlastLayer.js           Line-clear effect
    StageBanner.js          Stage-up banner
    ThemeSelector.js        Theme shop
    AdWatchModal.js         Rewarded-ad modal
    Icon.js                 Emoji-glyph icons (no lucide-react on native)
  engine/
    gameLogic.js            Placement, line detection, scoring
    shapes.js               Piece definitions
    soundEngine.js          Runtime WAV synth via expo-av + haptics
    storage.js              AsyncStorage persistence (cached, sync reads)
  styles/themes.js          Theme palettes
android/                    Native project (generated; safe to delete + prebuild)
```

Ignore the `.jsx` files

Some folders contain **stale `.jsx` twins** (`GameScreen.jsx` , `App.jsx` , ...) left from an early web prototype. They use `<div>` , `localStorage` , and `lucide-react` and **do not run on Android**. The build only uses the `.js` files.

Always edit the `.js` version. The `.jsx` files are dead weight and should eventually be deleted.

---

7. Troubleshooting — read this first

`ERROR: JAVA_HOME is set to an invalid directory`

`JAVA_HOME` points at a JDK that was uninstalled or moved. Repoint it at JDK 17 (§2.2) and open a new terminal.

`[CXX1210] No compatible library found`

Two very different causes:

(a) `JAVA_TOOL_OPTIONS` is set. This one is nasty and the error message is a lie. If that variable is set (e.g. `-Dlog4j2.formatMsgNoLookups=true`), the JVM prints `Picked up JAVA_TOOL_OPTIONS: ...` to stderr on every launch. Gradle reads prefab's stderr, sees unexpected text, and reports "No compatible library found" **even though prefab succeeded**. Check and clear it:

```
echo $JAVA_TOOL_OPTIONS          # should be empty
unset JAVA_TOOL_OPTIONS          # current shell (bash)
```

```
[Environment]::SetEnvironmentVariable("JAVA_TOOL_OPTIONS", $null, "User")  # permanent
```

(b) `node_modules` **was copied from another machine.** Fix:

```
rm -rf node_modules package-lock.json
npm install
npx expo prebuild --clean
```

`There is not enough space on the disk`

Android builds need several GB. Reclaim safely:

```
rm -rf ~/.gradle/caches/build-cache-1        # regenerates
rm -rf ~/.gradle/caches/transforms          # regenerates (often 2+ GB)
cd android && ./gradlew clean
```

Also delete unused emulator system images in Android Studio → Device Manager (each is ~2.5 GB).

`Unable to load script` / red screen

Metro isn't running or the phone can't reach it.

```
npx expo start          # keep this terminal OPEN -- closing it kills Metro
adb reverse tcp:8081 tcp:8081
```

`The required package 'expo-asset' cannot be found`

```
npx expo install expo-asset
```

App installs but immediately closes

Read the real error:

```
adb logcat -c        # clear
# relaunch the app, then:
adb logcat | grep -iE "ReactNativeJS|FATAL|AndroidRuntime"
```

Native module changes not appearing

You hot-reloaded when you needed a rebuild. Run `npx expo run:android`.

Nuclear reset (when all else fails)

```
rm -rf node_modules package-lock.json android
npm install
npx expo prebuild --clean
npx expo run:android
```

`android/` is generated — deleting it is safe **unless** you hand-edited native files, so check `git status` first.

---

8. Working agreements

- **Never commit** `node_modules/`, `android/app/build/`, `.expo/`, `*.apk`, or `android/local.properties`. The `.gitignore` covers these.
- **Never commit machine-specific paths.** No `org.gradle.java.home`, no `C:\Users\<you>\...` anywhere in tracked files.
- **Use `npx expo install`** for Expo/React Native packages.
- **Commit `package-lock.json`.** It keeps everyone on identical versions.
- After pulling changes that touch `package.json`, run `npm install`; if the change added a native module, also re-run `npx expo run:android`.
- Branch per feature, PR into `main`.

---

9. First-day checklist

```
[ ] Node 20 or 22 installed          node -v
[ ] JDK 17 installed                 java -version   -> 17.x
[ ] JAVA_HOME set                    echo $JAVA_HOME
[ ] JAVA_TOOL_OPTIONS empty          echo $JAVA_TOOL_OPTIONS
[ ] Android Studio installed
[ ] SDK 35 + 34, Build-Tools 35.0.0
[ ] NDK 26.1.10909125 installed      (exact version)
[ ] ANDROID_HOME set                 adb --version
[ ] 25 GB free disk
[ ] Repo cloned, npm install clean
[ ] Device shows in adb devices
[ ] npx expo run:android succeeds
[ ] Edited a JS file, saw hot reload
```